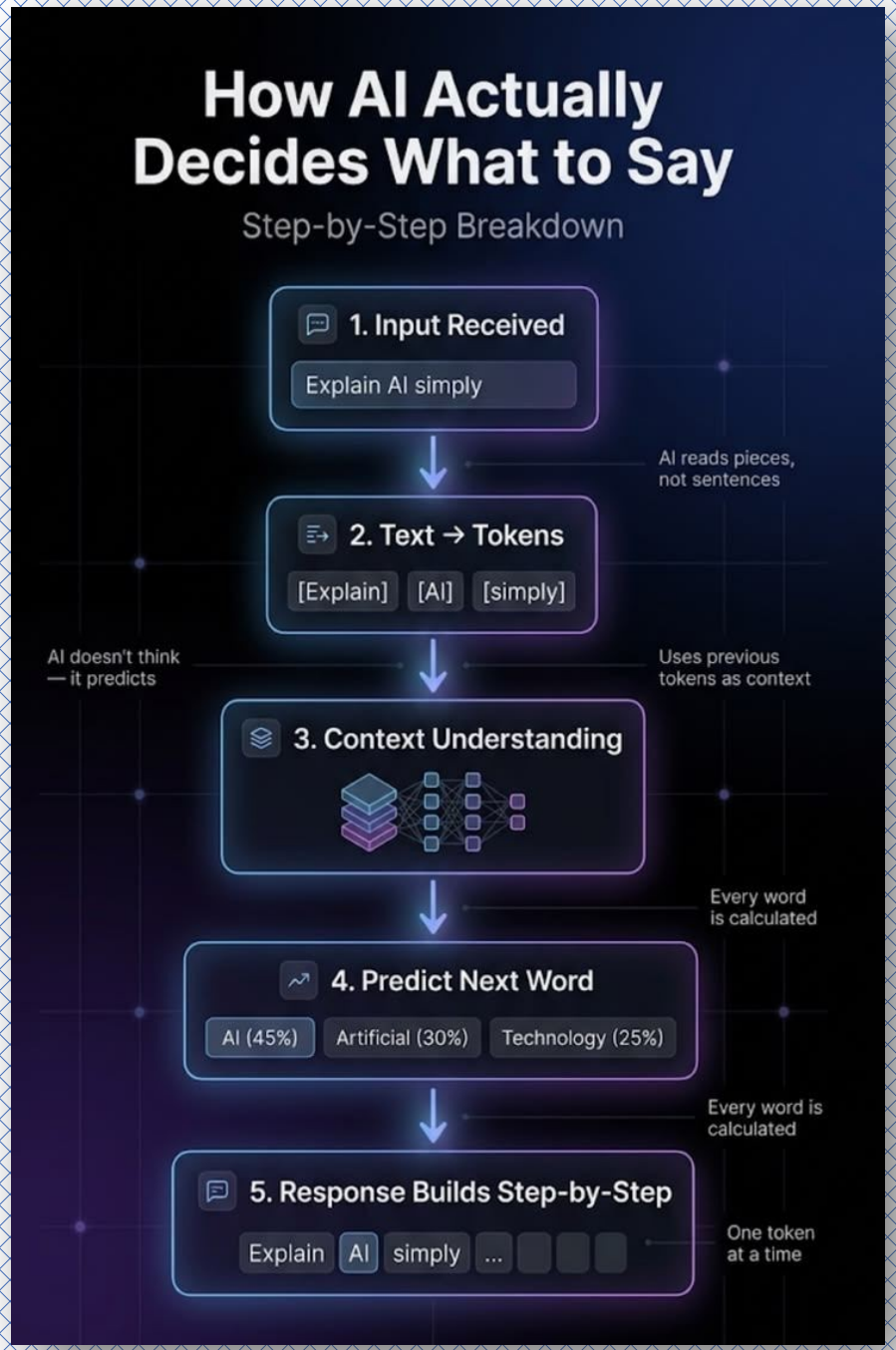
AI Prompting and Version Control Using Git

A Project-Based Approach



AI does not “think” like humans; it generates responses by predicting the next word (token).

The input text is first received and understood as raw data.

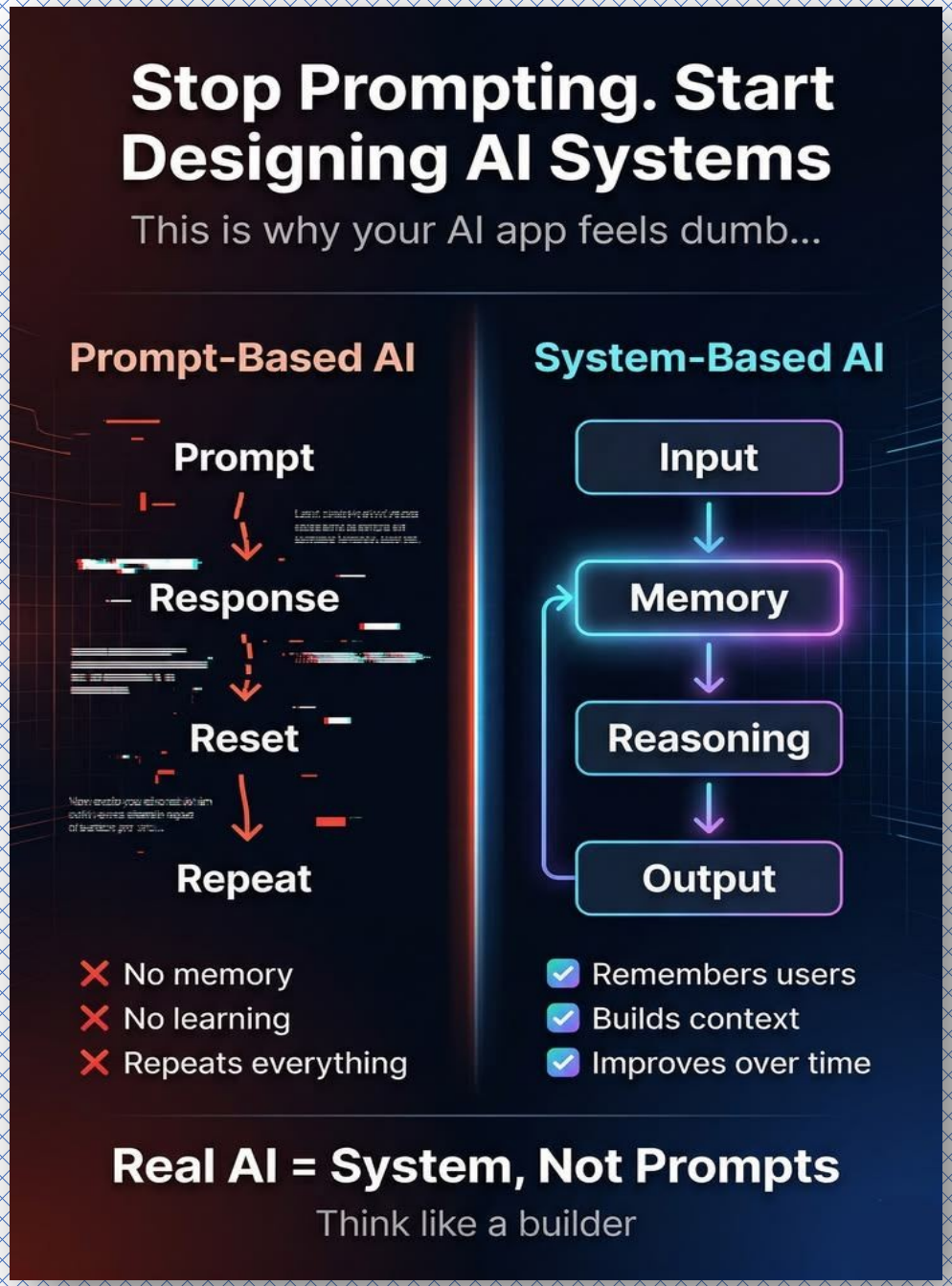
The sentence is broken into smaller units called tokens (words or sub-words).

AI uses context from previous tokens to understand meaning and relationships.

It then applies probability to predict the most likely next word.

This process repeats step-by-step until a full response is generated.

Each word is mathematically calculated, not consciously chosen.



Prompt-Based AI (Traditional Approach)

Works only based on single prompts and responses
No memory of previous interactions
Every request is treated as independent (reset after each response)
Cannot learn or improve from past interactions
Often leads to repeated or inconsistent outputs

System-Based AI (Modern Approach)

Designed as a complete AI system pipeline
Includes Input → Memory → Reasoning → Output

Maintains context across interactions
Can remember users and past data
Produces more consistent and intelligent responses
Improves over time through system design

Zero-Shot Prompting

* The model generates output without any example inputs.
* Useful when quick responses are needed without training data.
* Works well for general or simple tasks.
* Depends fully on the model's pre-trained knowledge.
* Best when you want fast and flexible results.

Few-Shot Prompting

* The model is given a few examples before generating output.
* Helps improve accuracy for specific tasks.
* Useful when task pattern needs to be learned from examples.
* Requires some sample data for better performance.
* More reliable for structured or domain-specific tasks.

GitHub VS git

1. It is a service
2. It is hosted on Web
3. It provides a graphical interface
4. It is a space to upload a copy of the Git repository
5. It provides functionalities of Git like VCS, Source Code Management as well as adding few of its own features

1. It is a software
2. It is installed locally on the system
3. It is a command line tool
4. It is a tool to manage different versions of edits, made to files in a git repository
5. It provides functionalities like Version Control System Source Code Management

A TINY CHEATSHEET ON

GIT

git is the most popular version control system. Being proficient in git is a must for every developer.

git init
Create an empty repository in the current directory

git add [file]
Add the **[file]**(s) to the *staging area*

git commit
Create a snapshot of all the files in the *staging area*

git push [name] [b]
Push changes to a remote repository called **[name]** to branch **[b]**

git pull [name] [b]
Pull any changes from a remote repository called **[name]** from branch **[b]**

git branch
List all of the branches in your repo. Add a **[branch]** argument to create a new branch called **[branch]** .

git remote add [name] [url]
Add a remote repository with **[url]** and an alias of **[name]**

git checkout -b [branch]
Switch to a branch named **[branch]**

git merge [branch]
Merge branch named **[branch]** with the current one

git clone [url]
Download a git repository from **[url]**

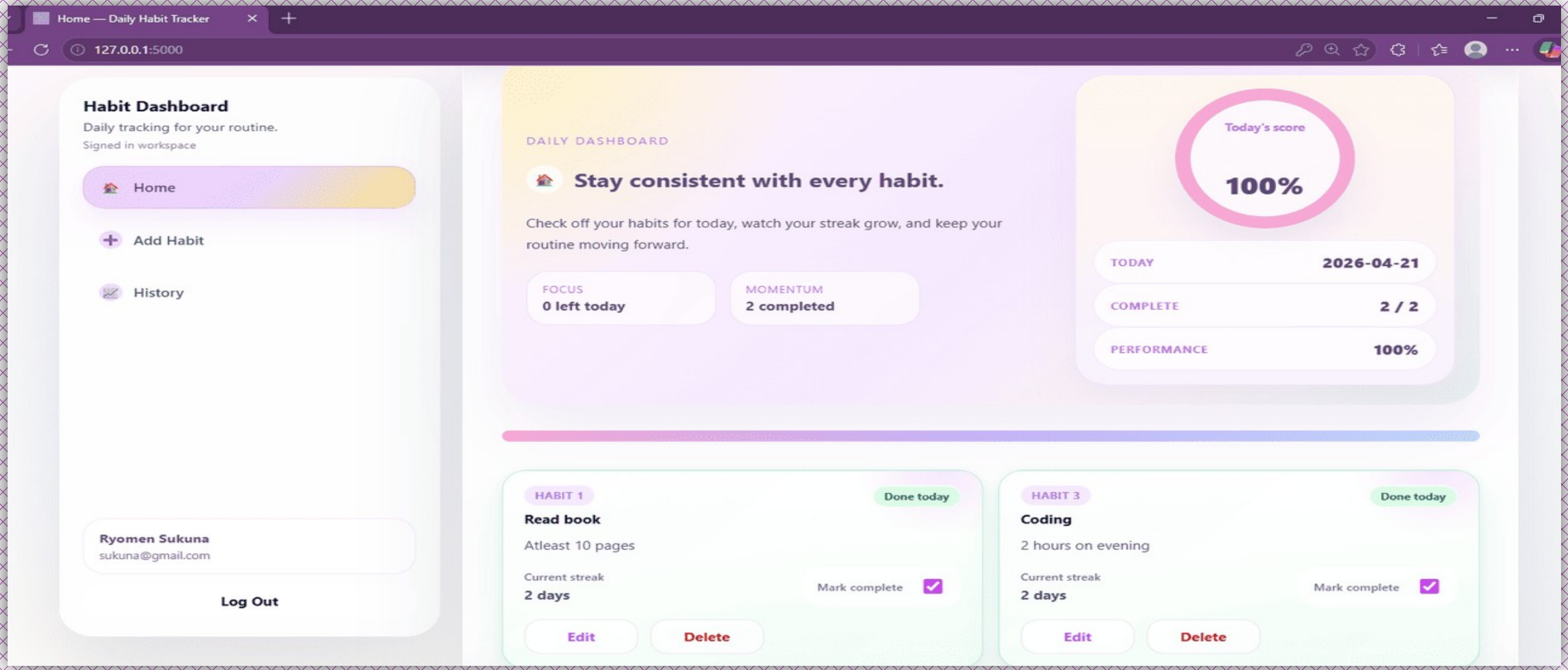
git log
List all the commits in the current branch's history

Daily Habit Tracker

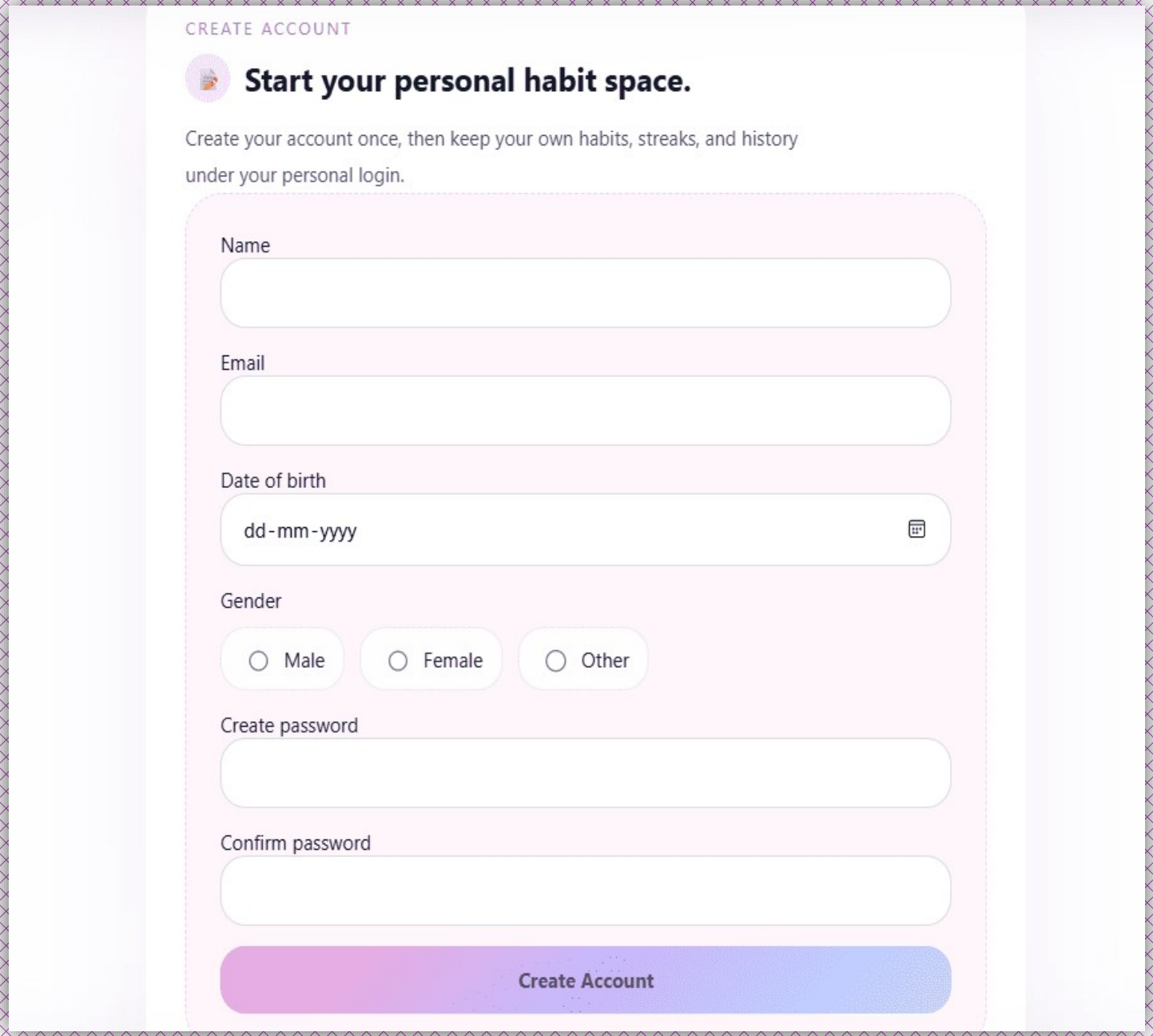
In today’s fast-paced lifestyle, maintaining consistency in daily routines has become a major challenge for many individuals. People often struggle to follow their goals due to lack of proper tracking, motivation, and accountability.

The **Daily Habit Tracker** is a web-based application designed to help users build and maintain positive habits in a structured and organized manner. It allows users to track their daily activities, monitor progress, and analyze their performance over time.

The system provides a personalized experience where each user can manage their own habits, mark daily completion, and view their consistency through streaks and progress indicators. By simplifying habit monitoring, the application encourages discipline and helps users achieve long-term personal goals through small daily improvements.



- It provides a complete overview of all active habits in one place.
- Users can view their daily habits and track their progress easily.
- Each habit is displayed with its current status (completed / not completed).
- Users can mark habits as completed directly from the dashboard.
- The page shows streak information, helping users track consistency.
- Progress indicators (like percentage or progress bars) are used for better visualization.



Signup Page

It allows new users to create an account in the *Daily Habit Tracker* application.

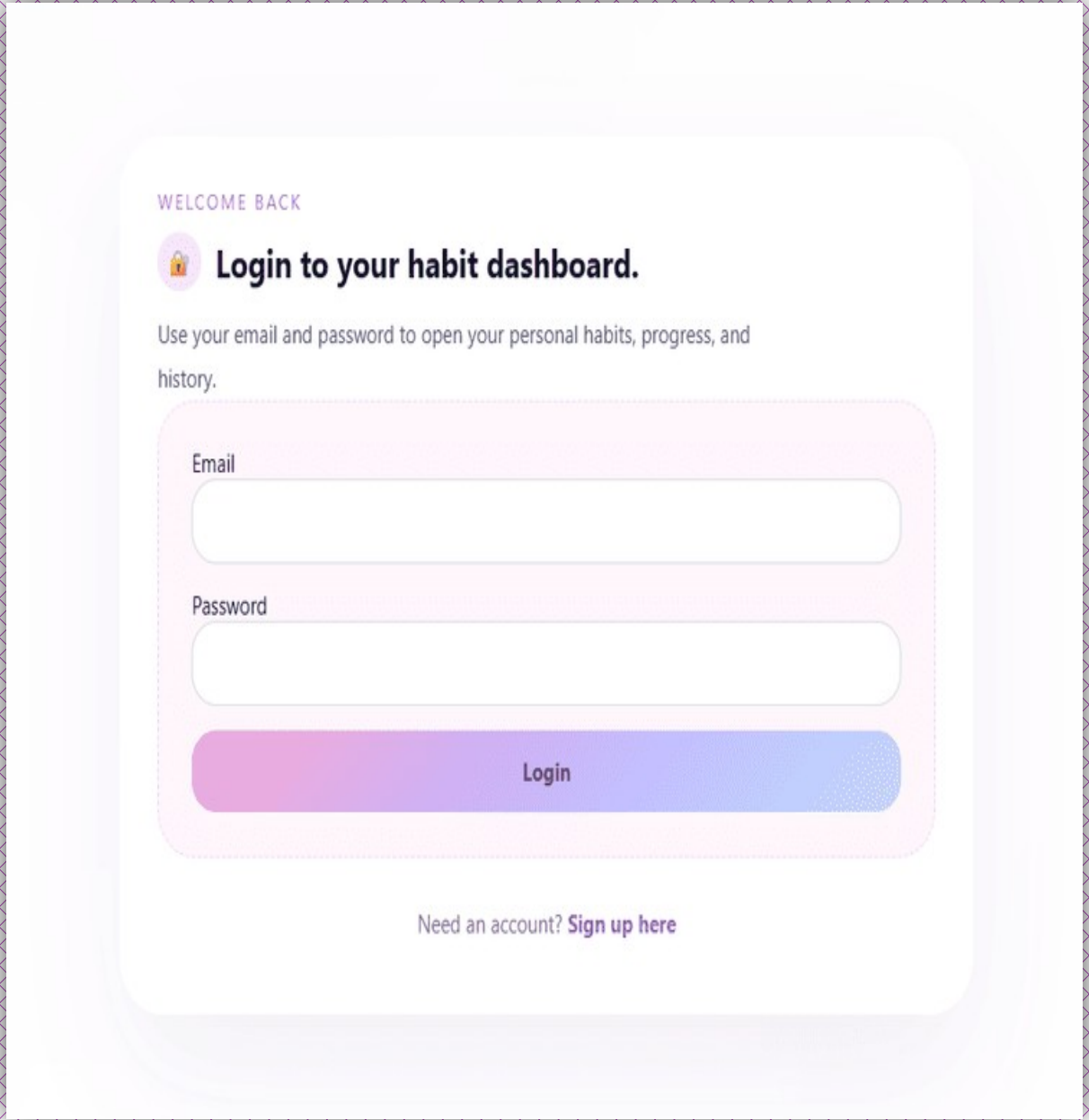
Users must provide basic details such as name, email, password, date of birth, and gender.

The system ensures secure registration by storing passwords in hashed format.

It checks for valid input fields and prevents duplicate email registrations.

Proper form validation is implemented to ensure correct user data entry.

After successful signup, the user is redirected to the login page.



Login Page

It allows existing users to access the Daily Habit Tracker system securely.

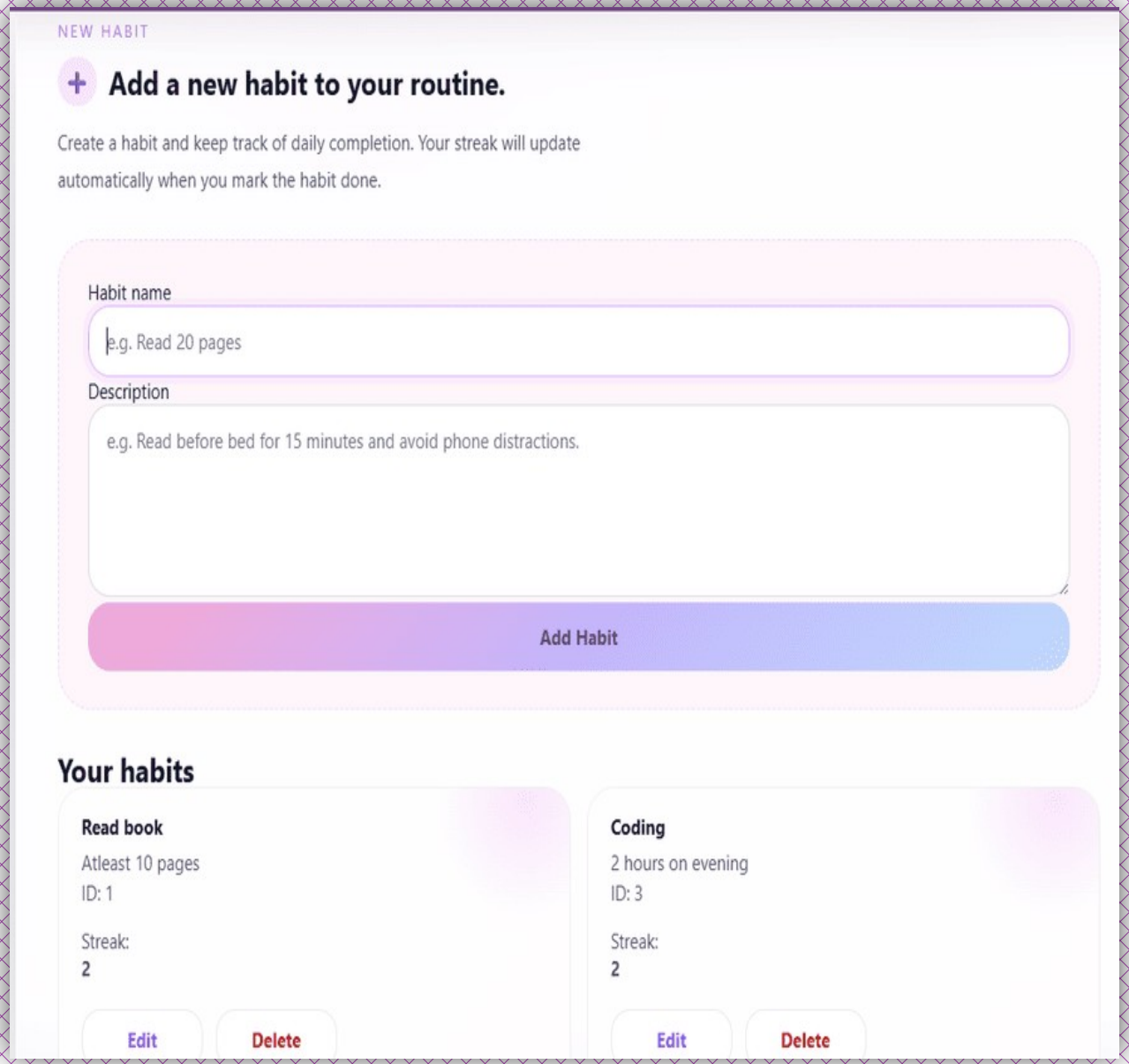
Users must enter registered email and password to log in.

The system performs credential verification using stored hashed passwords.

If the credentials are correct, the user is authenticated and redirected to the dashboard (home page).

If incorrect details are entered, the system shows appropriate error messages (invalid login).

It ensures secure access and prevents unauthorized entry into user data



Add Habit Page

It allows users to create new habits in the Daily Habit Tracker system.

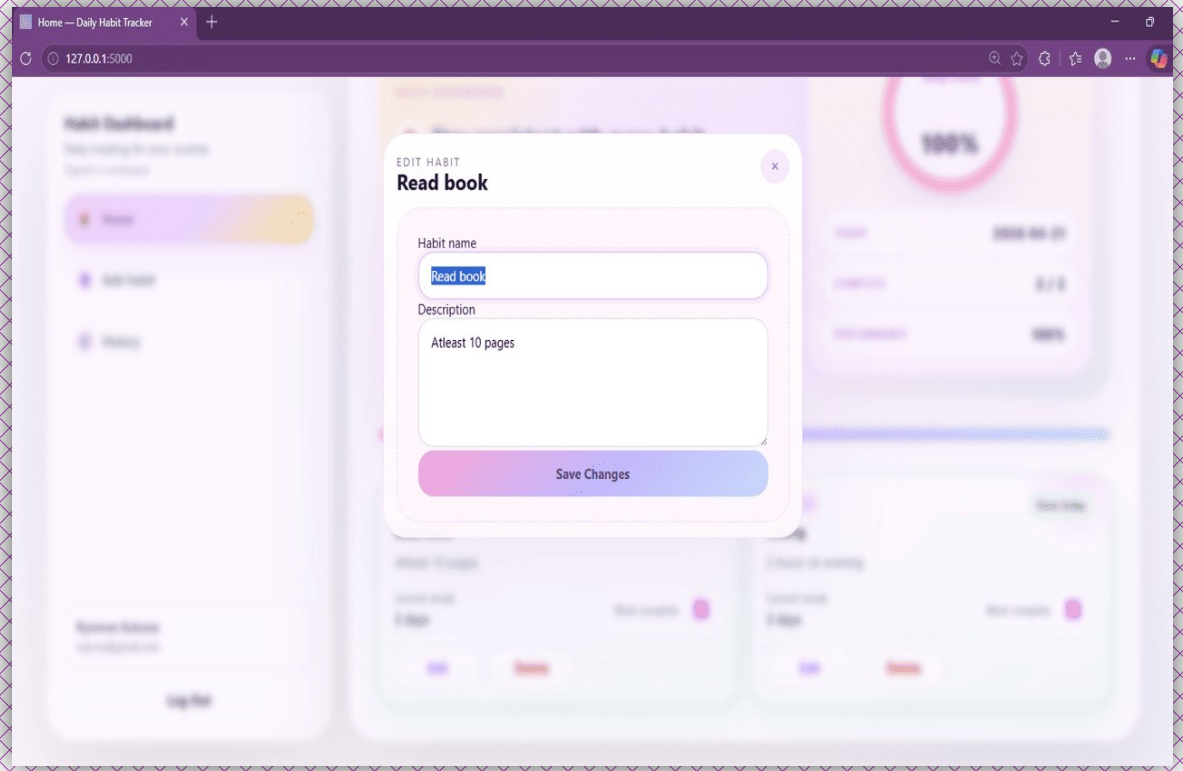
Users can enter habit name and description to define their daily activity.

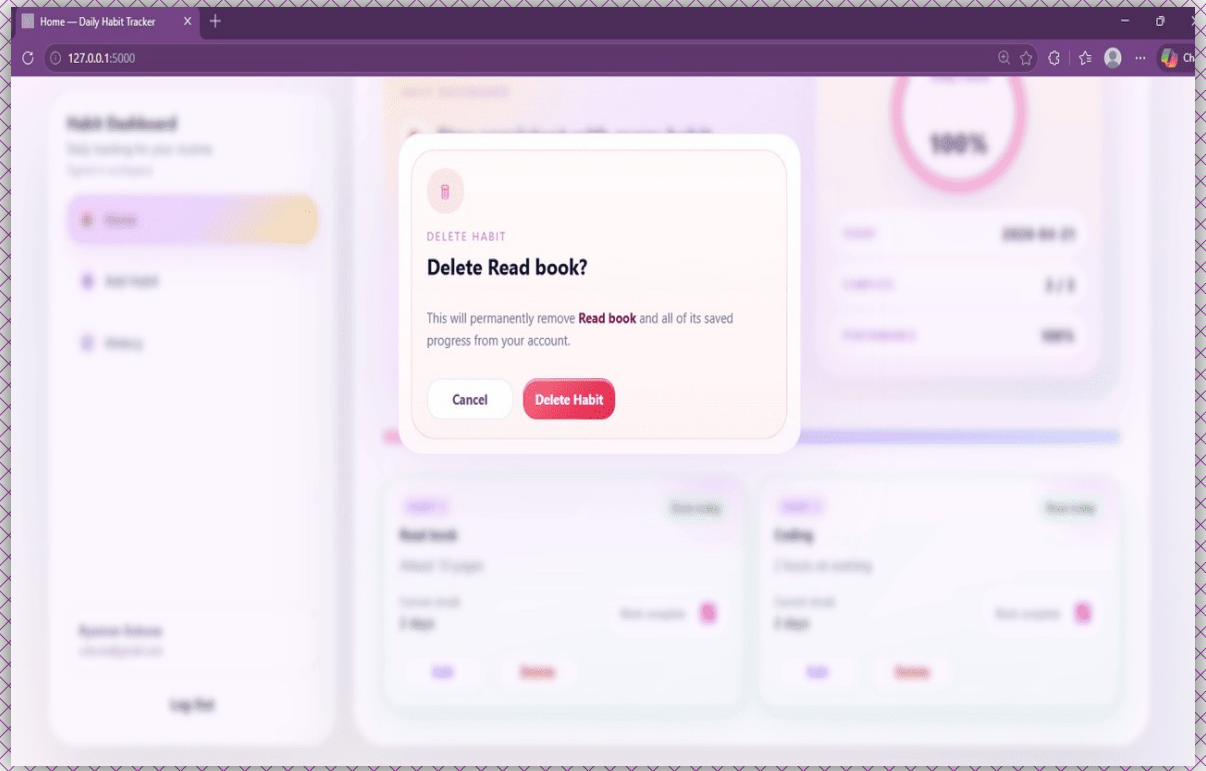
Each habit is assigned a unique ID for proper tracking and management.

The system validates input to ensure the habit name is not empty.

Once added, the habit is stored in the user’s personal habit list (JSON file).

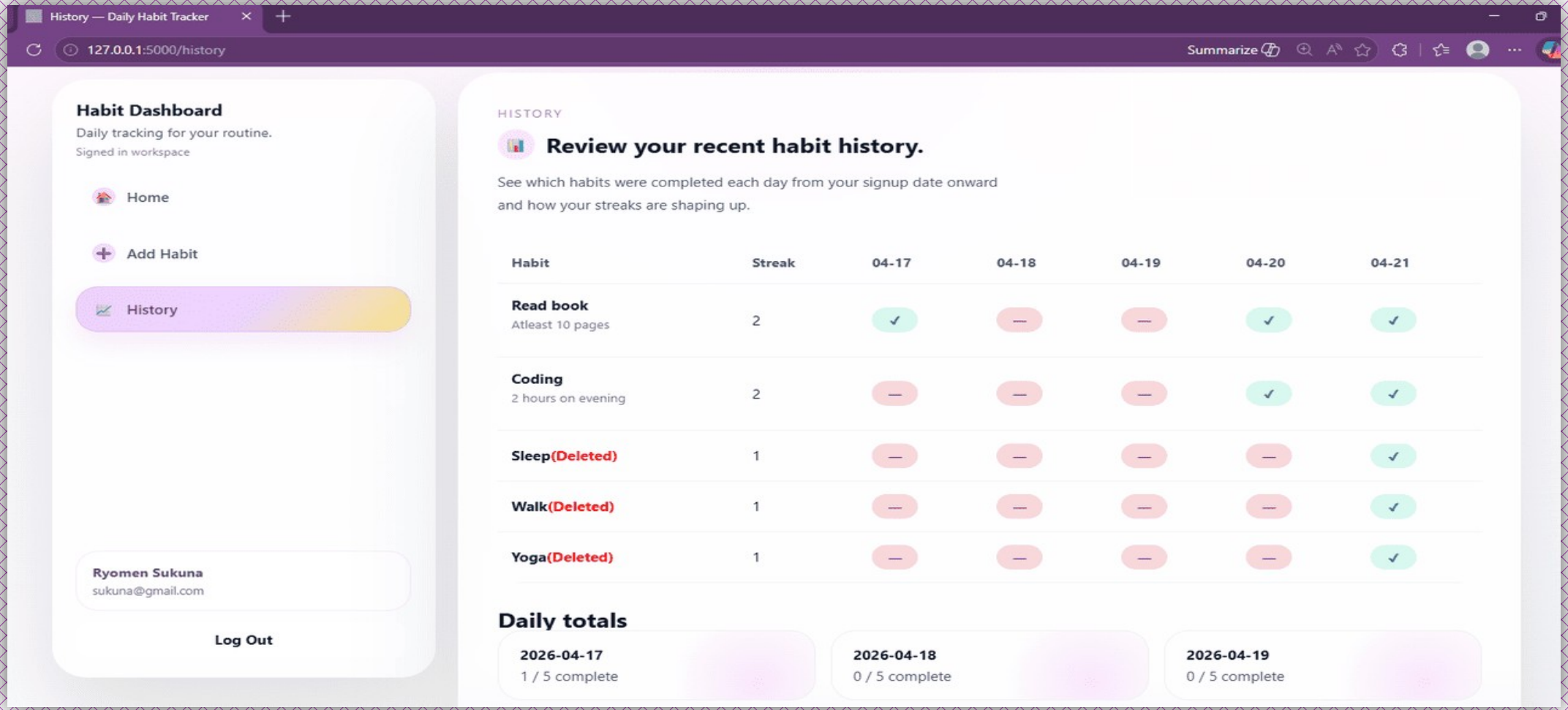
Newly added habits are immediately reflected in the dashboard.





Edit & Delete Popup (Modal)

* The Edit and Delete features are implemented using popup modals for better user experience.
* These popups allow users to modify or remove habits without leaving the current page.
* The Edit Popup enables users to update the habit name and description easily.
* Changes are instantly saved and reflected in the dashboard after submission.
* The Delete Popup provides a confirmation step before removing a habit.
* Instead of permanent deletion, the system uses soft delete (is_deleted = True) to maintain data history.
* This approach helps prevent accidental data loss.



History Page

* It displays the past records of user habits and their completion status.
* Users can analyze their consistency and performance over time.
* It helps in identifying completed and missed habits for each date.
* Even deleted habits are retained in history for complete tracking.
* This feature supports performance evaluation and self-improvement.

Thank you